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Investigating the Coupled Effects of Particle Size and Surface Oxidation on Metformin Adsorption Kinetics in Polyethylene Microplastics

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Research topic: surface chemistry adsorption kinetics of pharmaceutical contaminants on oxidized polyethene microplastics

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Introduction

By way of hydrophobic interactions, electrostatic attraction and hydrogen bonding, as well as van der Waals and π–π type forces, the surfaces of microplastics are able to sequester other pollutants and transport them in the water column [1], [2]. Polyethylene (PE) is perhaps the best known polymer in this regard. In their work, Razanajatovo et al. [3] demonstrated that PE microplastics will sorb sulfamethoxazole, propranolol and sertraline to varying degrees, a phenomenon they put down to the drugs' respective hydrophobicity and electrostatic properties.

Even with metformin, a hydrophilic compound that is for the most part excreted unchanged, there are measurable levels of adsorption onto the hydrophobic PE. Batch experiments by Guimarães et al. [4] on low-density PE microspheres revealed that metformin hydrochloride adsorbs quickly, following pseudo-first and pseudo-second-order kinetics.

There are two variables at play when it comes to pharmaceuticals and PE. Weathering is one; UV exposure introduces oxygen-containing groups to the surface of the PE, rendering it more polar and altering its ability to sorb pharmaceuticals [5]. This was borne out in 2026 when a study [6] found that PE microspheres left to age for 30 or 120 days had a greater capacity for metformin than unaged material (the maximum went from 3.54 to 7.79 mg/g). The aging process also put cracks in the particle surface, increasing the specific surface area by as much as 77 per cent. Then there is the matter of particle size and surface area, which are established factors in sorption [7]. Research on six polymers in two different size classes showed that of the pharmaceuticals tested, only small virgin particles of PE, polypropylene and PVC, and both sizes of polyamide would adsorb them; and here too, aging made for a marked increase in adsorption [8].

One would be hard pressed to find a study in the literature that has subjected metformin on PE to such a design, taking into account the kinetics of adsorption as well as the oxidation state and size of the particle. Study [6] did age a certain type of PE, and while [8] has seen to it that size and aging are varied, it was for other polymers and drugs. They have each proven significant, but no one has yet tested if the effect of a particular particle size is contingent on the level of surface oxidation.

Existing Literature and Gap

The Gap: In the literature I reviewed, I did not find a study that combines particle size, oxidation state, and adsorption kinetics in a single factorial design for metformin on PE. Both factors have been shown to matter on their own, and [8] varied both for other drugs and polymers. What is untested is whether they interact and how, meaning whether the effect of particle size depends on how oxidized the surface is. That interaction is the main focus of this proposal.

Research Question, Aim, and Hypotheses

Research question

How does particle size affect the adsorption kinetics of metformin onto polyethylene microplastics across different degrees of surface oxidation, and does surface oxidation modify this size effect?

Aim

To find out whether the effect of particle size on the rate and extent of metformin adsorption changes as the PE surface becomes more oxidized. This separates the physical effect of size from the chemical effect of oxidation, and tests whether they are actually independent.

Hypotheses

One mechanism, in my opinion, that is plausible: metformin is strongly basic, so at pH 7 it carries a positive charge. Oxidized PE gains carboxyl-type groups that are negatively charged at pH 7, which should attract cationic metformin and allow hydrogen bonding. This fits [6], where aged PE had a more acidic point of zero charge, meaning a more negative surface at a given pH. If this is the main driver, oxidation adds binding sites per unit area, while particle size controls how many of those sites are reachable and how fast.

- **H1 (size):** Smaller particles will show faster metformin adsorption.
- **H2 (oxidation):** Higher oxidation will increase adsorption capacity (qe).
- **H3 (interaction):** Surface oxidation will significantly modify the effect of particle size on metformin adsorption kinetics. The direction of this interaction will be determined from the data rather than assumed in advance.

The null hypothesis for H3 is that the size effect is the same at every oxidation level. I will test H3 with a two-sided test, so a result where oxidation reduces the size effect would also count as a real finding.

Methodology

This is a quantitative, controlled batch-adsorption study. I vary two properties of PE, particle size and surface oxidation, and track metformin removal over time to build a kinetic profile for each combination.

Stage 1: PE preparation

PE particles from a single source will be rinsed with ultra-pure water and sieved into three size fractions:

- Large: 500–1000 µm
- Medium: 250–500 µm
- Small: 106–250 µm

Sieving does not guarantee a uniform size class for irregular particles, so I will measure each fraction's real size distribution with optical microscopy or laser diffraction. I will use that distribution to estimate the specific surface area (m^2/g) for each fraction, so that I can later separate a pure surface-area effect from everything else that the size has changed.

Stage 2: Surface oxidation

Each size fraction will be aged in an accelerated-weathering UV chamber at a fixed, reported irradiance, temperature, and humidity. Particles will be spread in thin layers and turned regularly so all of them are exposed evenly. Exposure times are:

- 0 h: unoxidized reference
- 100 h: intermediate oxidation
- 300 h: high oxidation

I will not assume the oxidation level from exposure time alone. Each batch will be checked with:

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- **ATR-FTIR carbonyl index:** the ratio of the carbonyl peak (~1715 cm^{-1}) to a stable reference peak (~1465 cm^{-1}), the standard proxy for PE oxidation.
- **Water contact angle:** measured on PE film pieces aged alongside the particles, since contact angle is hard to measure on loose powder. Oxidized PE should become more hydrophilic.
- **XPS O/C ratio** (if instrument access allows): to confirm oxygen-containing groups at the surface.

The 0, 100, and 300 h labels represent nominal UV-aging treatments, not assuming oxidation states. UV aging can also make PE brittle and crack on its surface, which, in [6], raised specific surface area by up to 77%. So I will re-check the size distribution after aging and re-estimate surface area (using BET or krypton adsorption if available), because a size measurement alone would miss the surface cracking. If the carbonyl indices of the 100 h and 300 h samples turn out too close to separate, I will analyze oxidation using the measured carbonyl index as a continuous variable instead of three fixed groups (see Data Analysis).

Stage 3: Adsorption experiments

Each of the nine size × oxidation combinations will be run under identical conditions:

- PE dose: 1 g/L
- Initial metformin concentration: 5 mg/L
- pH 7, held with low-UV-absorbance buffer (e.g., phosphate) at fixed ionic strength
- 25 °C in a controlled shaking incubator, with the same agitation speed for every condition
- Sealed vessels with the same solution volume, and particles pre-wetted so that floating, hydrophobic PE is properly dispersed

Remaining metformin will be measured at 0, 5, 15, 30, 60, 120, 240, and 1440 min. Sampling is weighted toward early times because adsorption of metformin onto PE has been reported to be relatively fast [4]. The 240 and 1440 min points will show whether equilibrium has been reached. Each sample will be a small aliquot (total removed volume kept under about 10%, and corrected for in the calculations), separated from the particles by centrifugation or by a filter that I have first tested for metformin loss.

Pilot run: With hydrophilic metformin, PE particles this large, and a 1 g/L dose, the drop in concentration could be small. Before the main experiment, I will run one pilot condition to check that the drop is clearly larger than the measurement error. If it is not, I will raise the PE dose or the metformin concentration, and keep the new value fixed across all nine conditions. Because capacity per gram can fall as PE dose rises [4], the dose I settle on has to be the same everywhere.

Stage 4: Kinetic analysis

Quantification

Metformin will be measured by UV-Vis spectrophotometry at 233 nm, using a calibration curve prepared in the same buffer (reporting the calibration range, R^2, and detection limit) and checked with spike-recovery tests to verify the measured 233 nm signal tracks metformin quantitatively. Aged PE can release UV-absorbing degradation products, so I will run a metformin-free leachate blank for every size × oxidation combination and subtract it. Where instrument access allows, HPLC-UV will be used instead because it is more specific and less affected by leachates.

Quality control

- **PE-free blank:** checks for metformin lost to vessel walls, filters, or the buffer rather than to PE.
- **Metformin-free blanks** (one per size × oxidation combination): check for leachates that interfere with detection.
- **Triplicates:** every one of the nine conditions is run three times (n = 3), and run order is randomized so that no condition is always measured first or last.

Feasibility

I have not yet secured a lab partner, though my plan is to run the work under the teacher at my school and to approach a university or research-institute lab for FTIR and, if possible, HPLC. UV-Vis is the primary method for that reason, with HPLC-UV as an upgrade. If FTIR is not available, I will describe oxidation by UV exposure dose and by simpler measures such as contact angle, compare the trend with the oxygen-containing groups reported in [6], and state clearly in the results that oxidation was not directly verified.

Data Analysis

2 The amount of metformin adsorbed at time t is qt = (C0 − Ct) × V / m, where C0 and Ct are the initial and remaining concentrations, V is the solution volume, and m is the PE mass. Adjustments are made for volume removed by sampling and for blanks.

14 **Kinetic models:** I will fit pseudo-first-order and pseudo-second-order models, and use Weber–Morris intraparticle-diffusion plots to look for more than one linear segment instead of assuming a single rate-limiting step. Fitting will use nonlinear regression, not the linearized forms, because linearization can bias which model looks best. Models will be compared with AICc (a small-sample version of AIC) alongside R^2, and I will also ask which model makes physical sense, not only which has the highest R^2.

15 **Parameters compared:** For each replicate, I will extract the equilibrium capacity (qe), the rate constant (k), and the initial adsorption rate. I will also report qe and initial rate normalized by specific surface area. If the size effect disappears after normalizing, it is purely a surface-area effect. If it remains, something else about size are also involved.

Statistical tests: The three main comparisons are:

- particle size versus adsorption rate
- oxidation state versus adsorption rate
- the **particle size × oxidation interaction**, which is the most important because it addresses the gap

These will be tested with a two-way ANOVA on k and qe ($\alpha = 0.05$, Holm correction across responses), after checking normality and equal variance of residuals (Shapiro–Wilk, Levene) and transforming the data if needed. If the interaction is significant, I will follow up with simple-effects comparisons of size at each oxidation level. As a second analysis, I will regress the responses on measured carbonyl index and size, which does not depend on the three oxidation groups being cleanly separated.

2 With n = 3 per condition (18 error degrees of freedom), this design can only detect fairly large effects reliably. For that reason, I will report effect sizes and 95% confidence intervals alongside p-values, and will not treat a non-significant interaction as proof that no interaction exists.

Research Design

A 3 × 3 full factorial design: three size fractions across three oxidation levels, each combination run in triplicate (27 adsorption runs), plus blanks.

Particle size	0 h (unoxidized)	100 h (intermediate)	300 h (high)
Small (106–250 µm)	n = 3	n = 3	n = 3
Medium (250–500 µm)	n = 3	n = 3	n = 3
Large (500–1000 µm)	n = 3	n = 3	n = 3

The following are held constant across all nine conditions: pH, ionic strength, temperature, PE mass, initial metformin concentration, solution volume, agitation, vessel type, and sampling schedule. That way, differences in the results can be attributed to particle size and oxidation rather than to something else.

Expected Results

I expect smaller particles to adsorb metformin faster (more accessible area and shorter diffusion distances), and more oxidized particles to reach higher capacity (more oxygen-containing sites). I do not assume the direction of the size × oxidation interaction in advance. There are three ways the results could go, and each would mean something different:

- **Size effect grows with oxidation:** surface chemistry and surface area work together, which supports H3.
- **Size effect shrinks with oxidation:** chemistry dominates, and area matters less once sites are plentiful.
- **No interaction:** the two effects are independent, and single-factor studies are a reasonable approximation.

All three outcomes would improve on the current single-factor picture, so the study is informative regardless of which one appears.

Risk Management

Risk	How I will manage it
Uneven size separation	Measure the real size distribution of every fraction, before and after aging.
Variable or overlapping oxidation levels	Verify with carbonyl index and contact angle for every batch; fall back to carbonyl index as a continuous variable.
Metformin drop too small to measure	Pilot run first; adjust dose or concentration and keep it fixed across conditions.
Leachates interfering at 233 nm	Leachate blank for every condition; HPLC-UV where possible.

Metformin lost to vessels or filters	PE-free blanks; test filters before use, or centrifuge instead.
Aging changes surface area as well as chemistry	Re-measure size and surface area after aging, and report area-normalized results.
Floating or aggregating particles	Pre-wet particles, use sealed vessels with fixed agitation, and note that effective size may differ from nominal size.
Random error	Triplicates, randomized run order, and calibration checks with spike recovery.

Because the study handles a pharmaceutical compound, all work will follow the institution's chemical-safety procedures under appropriate supervision.

Limitations and Future Work

- **Concentration.** 5 mg/L is much higher than metformin typically found in natural waters (usually in the ng/L to µg/L range). I chose it so the concentration change is measurable with UV-Vis, and it is best seen as a mechanistic study concentration, not a realistic exposure level. A follow-up at µg/L levels would need LC-MS.
- **Lab aging versus real weathering.** UV in a chamber does not include abrasion, biofilms, or dissolved organic matter, so it will not fully reproduce natural weathering.
- **Nominal versus effective size.** Particles may aggregate, so effective size may differ from sieve size.
- **Narrow scope.** One polymer, one drug, one pH, and a 24-hour window. Holding these fixed is what lets differences be attributed to size and oxidation, but it means the results cannot be assumed to apply directly to natural water.

Natural next steps are adsorption isotherms, other pH values and ionic strengths, desorption experiments, other polymers, and lower concentrations.

Significance and Potential Impact

The value of this work for environmental risk assessment is not just a matter of filling in some gaps in the literature. Consider microplastics out in the field: one will find them in all manner of sizes and states of weathering. Most laboratory studies do not reflect this, preferring to work with a single size and condition. If there is an interplay between oxidation and size, then any risk model or water-treatment approach that treats them as separate entities is liable to misjudge how much pharmaceutical load these particles are carrying.

Conclusion

It has been established by earlier research that PE can adsorb pharmaceuticals; metformin is a case in point, doing so in a measurable fashion that fits kinetic models. We also know aging changes the sorptive properties of the PE surface. But the question remains as to how particle size and surface oxidation work in tandem. This proposal is designed to put that to rest. Through a 3 × 3 factorial experiment, we will test and verify every variable rather than make assumptions, answering head-on whether the two factors interact.

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